

Automated Multi-Controller Batch Benchmark Engine, Empirical Comparative Evaluation, and Master Hybrid Adaptive Solver for a 6-Bar Tensegrity Icosahedron Planetary Rover

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Abstract— Compliant tensegrity rovers offer unmatched impact survivability and terrain adaptability for planetary surface missions, yet their complex underactuated mechanics make controller selection highly terrain-dependent. This paper introduces an end-to-end automated batch benchmark engine and a novel Master Hybrid Adaptive Solver for a high-fidelity 6-bar, 24-cable tensegrity icosahedron rover with an internally suspended 0.1D scientific payload core. Operating inside a 500 Hz semi-implicit integration physics environment, the headless benchmark systematically evaluates 14 distinct locomotion controllers across 14 procedurally generated Martian terrain stress profiles (196 full test runs). Controllers are evaluated using a multi-objective Composite Performance Score (CPS) that balances forward speed, distance progress, mechanical energy cost, peak payload shock (G-force), structural deformation, and cable safety. To overcome the performance trade-offs of individual controllers, we design and synthesize the Master Hybrid Adaptive Solver: a supervisory finite-state machine that dynamically switches between constrained Quadratic Programming Model Predictive Control (QP-MPC), robust iterative Linear Quadratic Regulators (iLQR Minimax), and learned adversarial policies based on real-time terrain telemetry. Discontinuous tension shocks during mode transitions are eliminated via a C^2 -smooth cubic Hermite spline actuation handover mechanism. The hybrid solver achieves the highest aggregate score across all terrain categories ($CPS = 2.14$), outperforming all standalone solvers while maintaining payload shock strictly below 1.48 G and structural deformation under 0.052 m. Complete telemetry streaming pipelines export goal-tracking, formation health, individual cable stresses, and contact dynamics to CSV and JSON formats for rigorous reproducibility.

Index Terms— Tensegrity Robotics, Planetary Exploration Rovers, Model Predictive Control (MPC), Iterative LQR, Supervisory Hybrid Control, Cubic Hermite Actuation Blending, Payload Shock Isolation, Headless Batch Benchmarking.

I. INTRODUCTION

Planetary exploration rovers operating on extraterrestrial surfaces such as Mars, the Moon, and Titan face extreme mechanical hazards. Traditional exploration rovers—such as the Mars Science Laboratory (Curiosity) and Mars 2020 (Perseverance)—rely on rigid aluminum chassis connected to wheeled rocker-bogie suspension systems. While exceptionally capable on gently sloping regolith, rigid rovers carry fundamental vulnerabilities: excessive structural mass, risk of joint actuator seizure from atmospheric dust, and catastrophic rollover vulnerability on steep crater walls or fields of jagged monolithic boulders [1].

What is a Tensegrity Rover? In simple terms, a *tensegrity* (tensional integrity) rover replaces rigid chassis and articulated axles with a continuous web of elastic tension cables suspending a constellation of isolated rigid compression struts [2]. In the classic 6-bar icosahedron architecture evaluated in this work, the 6 compression struts never touch each other; instead, they “float” within a protective 24-cable network (Fig. 1). When the rover drops off a 2-meter drop or strikes a massive rock, the impact shock is not concentrated at a vulnerable hinge. Instead, the force

is distributed instantly and omnidirectionally across the entire 3D cable lattice as elastic potential energy. The entire robot functions simultaneously as its own wheel, suspension, roll-cage, and impact decelerator, making structural breakage nearly impossible.

Core Engineering Dilemma: Because a tensegrity rover has no rigid wheels, it achieves locomotion by actuating cable rest lengths to shift its Center of Mass (COM) outside its current triangular support base, causing it to tumble or roll forward onto an adjacent triangular face. However, this produces a highly nonlinear, non-smooth, underactuated dynamical system. Different control paradigms exhibit extreme performance disparities across varied terrains: an algorithm optimized for fast cruising on flat ground often fails catastrophically when attempting to climb an 18° boulder slope or negotiate low-friction marsh.

Prior literature has explored individual tensegrity control techniques—including Linear Quadratic Regulators (LQR) [2], Iterative LQR (iLQR) [3], Model Predictive Control (QP-MPC) [3], and Central Pattern Generators (CPGs) [4]. However, there has been no comprehensive empirical evaluation comparing these distinct control paradigms across an exhaustive suite of extraterrestrial terrains under identical physical conditions.

Furthermore, existing controllers operate as fixed standalone policies, failing to adapt when the robot transitions from flat regolith into jagged rock fields or high-slip mud bogs.

This paper resolves these limitations through five primary scientific contributions:

1. **Headless Batch Benchmark Engine:** A 500 Hz decoupled simulation engine that executes automated 14×14 matrix experiments across 14 controllers and 14 terrain levels at maximum CPU throughput.
2. **Comprehensive Multi-Controller Portfolio:** Rigorous mathematical implementation of 14 distinct controllers spanning linear feedback, iterative trajectory optimization, constrained receding-horizon optimization, and neural policies.
3. **Procedural Martian Terrain Benchmark:** 14 procedurally generated Martian test tracks categorized into four functional stress tests: Flat Transit, Rock Fields, Steep Slopes, and High-Slip Wetlands.
4. **Master Hybrid Adaptive Solver:** A terrain-aware supervisory controller that selects optimal policies in real time and utilizes a C^2 cubic Hermite spline blender to

guarantee zero actuation shock during mode transitions.

5. **Multi-Objective Composite Performance Score (CPS):** An empirical ranking formulation that objectively scores controllers across speed, energy, payload isolation, and structural safety.

II. SYSTEM TOPOLOGY AND PHYSICS FORMULATION

A. Hardware Geometry and Mass Distribution

The rover geometry follows the expanded 6-bar tensegrity icosahedron shown in Fig. 1. The structure consists of 6 rigid cylindrical compression rods ($j \in \{1, \dots, 6\}$), 12 outer node vertices ($\mathbf{p}_1, \dots, \mathbf{p}_{12} \in \mathbb{R}^3$), and 24 outer tensile actuation cables ($c_1, \dots, c_{24}$). Suspended at the geometric centroid is a spherical instrument core of diameter $D_{\text{core}} = 0.1D$ (0.10 m), constrained by 8 internal elastic suspension tethers to protect sensitive optical and scientific instrumentation.

TABLE I: ROVER TOPOLOGY, MECHANICAL PROPERTIES, AND PHYSICAL SIMULATION CONSTANTS

PARAMETER CATEGORY	SYMBOL	VALUE	UNIT	PHYSICAL MEANING / SYSTEM FUNCTION
Outer Rover Diameter	D	1.00	m	Outer envelope diameter measured between opposing node pairs
Strut Dimensions	$L_{\text{rod}}, r_{\text{rod}}$	1.00, 0.015	m	Length and outer radius of the 6 rigid carbon-fiber compression rods
Payload Core Diameter	D_{core}	0.10	m	Diameter of centrally isolated scientific payload instrument package (0.1D)
Mass Breakdown	$m_{\text{node}}, m_{\text{core}}$	0.20, 1.60	kg	12 outer nodes (2.4 kg) + central core (1.6 kg) = 4.00 kg total rover mass
Nominal Rover Weight	$W_{\text{Earth}}, W_{\text{Mars}}$	39.24, 14.88	N	Total gravitational downforce on Earth (9.81 m/s ²) and Mars (3.72 m/s ²)
Cable Rest Pretension	T_0	40.0	N	Baseline internal pretension maintaining icosahedron structural equilibrium
Cable Elastic Stiffness	k_s	1200.0	N/m	Linear tensile spring constant of the 24 outer Dyneema actuation cables
Internal Core Stiffness	k_{core}	1600.0	N/m	Spring stiffness of the 8 internal suspension tethers securing the scientific core
Actuator Rest Length Stroke	ΔL_{max}	± 0.120	m	Maximum motor spool displacement authority (± 12 cm) from baseline
Motor Spool Rate Limit	v_{spool}	0.40	m/s	Maximum linear line-retraction velocity of onboard miniature DC winches
Cable Overload Breaking Limit	T_{max}	700.0	N	Critical tension threshold; exceeding this triggers a structural failure penalty
Ground Contact Stiffness	k_g	40,000	N/m	Hertzian compressive contact penalty stiffness against terrain surfaces
Coulomb Surface Friction	μ_g	0.85	—	Baseline dry regolith coefficient of friction (scaled to 0.15 on marsh terrains)
Physics Integration Time Step	Δt_{phys}	0.002	s	500 Hz fixed-step numerical integration rate ensuring numerical stability
Control Decision Time Step	Δt_{ctrl}	0.020	s	50 Hz discrete controller rate (10 physics integration substeps per control tick)

Table I Explanation: System configuration parameters for the 6-bar tensegrity icosahedron rover. The physical parameters mirror physical prototypes developed in NASA Ames SUPERball research [1], with scaled masses and motor spool constraints corresponding to deployable planetary lander hardware.

B. Equations of Motion and Rigid Rod Constraint Projection

Let $\mathbf{q} = [\mathbf{p}_1^T, \dots, \mathbf{p}_{12}^T, \mathbf{p}_{\text{core}}^T]^T \in \mathbb{R}^{39}$ represent the generalized

coordinate vector of the 12 outer nodes and the central payload core. The unconstrained equations of motion are governed by Newton-Euler mechanics:

$$\mathbf{M} \ddot{\mathbf{q}}(t) = \mathbf{F}_{cable}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{F}_{ground}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{F}_{grav} + \mathbf{F}_{rigid} \quad (1)$$

where $\mathbf{M} \in \mathbb{R}^{39 \times 39}$ is the diagonal mass matrix, $\mathbf{F}_{cable}$ is the vector of internal cable forces, $\mathbf{F}_{ground}$ accounts for terrain contact and friction, $\mathbf{F}_{grav} = \mathbf{M} [0, 0, -g]^T$ represents planetary gravity, and $\mathbf{F}_{rigid}$ represents internal constraint forces that enforce the rigidity of the 6 compression struts.

Rigid Strut Enforcement via SHAKE: Each of the 6 carbon-fiber rods connects a pair of nodes (i, j) separated by nominal length $L_{rod} = 1.0$ m. To prevent rods from stretching or compressing without incurring numerical stiffness from artificial stiff spring penalties, we enforce the holonomic geometric constraint:

$$\sigma_k(\mathbf{q}) = \|\mathbf{p}_i - \mathbf{p}_j\|^2 - L_{rod}^2 = 0, \quad \forall k \in \{1, \dots, 6\} \quad (2)$$

During each 500 Hz integration substep, positions are first integrated via semi-implicit Euler integration, and the SHAKE constraint projection algorithm iteratively projects node positions along the rod axis until the residual error $|\sigma_k| < 10^{-6}$ m, rigorously preserving rod rigidity.

C. Cable Force Laws: Standard Hookean vs. Adaptive Hermite Relaxation

Each cable $i \in \{1, \dots, 24\}$ connects two nodes with Euclidean distance $l_i = \|\mathbf{p}_a - \mathbf{p}_b\|$ and rest length $l_{0,i}(t) = L_{base} + \Delta L_i(t)$, where $\Delta L_i \in [-0.12, +0.12]$ m is commanded by the active locomotion controller.

1) Standard Linear Hookean Model (Model A): Because cables cannot push, cable tension is strictly non-negative (unilateral constraint):

$$T_i = \max(0, k_s (l_i - l_{0,i}) + d_s \dot{l}_i) \quad (3)$$

While mathematically straightforward, Equation (3) possesses a dangerous physical flaw: during aggressive obstacle impacts or rapid rest-length changes, the cable snaps from slack ($T_i = 0$) to

violent tension ($T_i > 600$ N) almost instantaneously. This high dT/dt impulse shock travels directly into the structure, exciting high-frequency oscillations that violently shake the internal payload core.

2) Adaptive Hermite Relaxation Law (Model B): To eliminate impact shock spikes, we formulate an adaptive nonlinear relaxation law. When cable strain $\varepsilon_i = (l_i - l_{0,i}) / l_{0,i}$ or strain rate $\dot{\varepsilon}_i$ spikes during dynamic collision events, an adaptive damping-relaxation factor $\psi(\varepsilon, \dot{\varepsilon})$ smoothly scales tension using a C^1 -smooth cubic Hermite function:

$$T_{i,relaxed} = T_0 + (T_i - T_0) \cdot [1 - \psi(|\dot{l}_i| / v_{threshold})] \quad (4)$$

This relaxation law prevents tension spikes from exceeding safe structural bounds, allowing the outer cage to deform elastically and absorb 88% of obstacle impact energy before it reaches the payload core.

D. Ground Contact Mechanics

When any node $\mathbf{p}_i$ penetrates the terrain elevation surface $z_{terrain}(x_i, y_i)$ with penetration depth $\delta_i = \max(0, z_{terrain} - z_i)$, a compressive Hertzian normal contact force is generated:

$$F_{n,i} = k_g \delta_i^{1.5} + d_g \delta_i^{0.5} \dot{\delta}_i \quad (5)$$

Tangential friction is computed using regularized Coulomb friction with static stiction: $\mathbf{F}_{t,i} = -\mu_g F_{n,i} \cdot \mathbf{v}_{t,i} / (\|\mathbf{v}_{t,i}\| + \varepsilon_v)$, where $\varepsilon_v = 0.01$ m/s avoids numerical discontinuity at zero velocity.

III. COMPREHENSIVE CONTROLLER TAXONOMY

The benchmark evaluates 14 distinct controllers spanning four core control paradigms. Table II summarizes their algorithmic characteristics, while detailed mathematical descriptions follow.

TABLE II: ALGORITHMIC CHARACTERISTICS AND COMPUTATIONAL PROFILES OF THE 14 LOCOMOTION CONTROLLERS

CONTROLLER KEY	DESCRIPTIVE ALGORITHMIC TITLE	CONTROL PARADIGM	UPDATE RATE	OPTIMIZATION OBJECTIVE / MATHEMATICAL STRATEGY	PRIMARY STRENGTH
<code>natural_support_face</code>	Natural Support-Face Gait	Geometric FSM	10 Hz	Heuristic triangle tipping via COM centroid projection	Simple, zero matrix inversion
<code>riccati_lqr</code>	LQR • Riccati Backward Pass	Linear Quadratic	50 Hz	Infinite-horizon discrete algebraic Riccati equation (DARE)	Global asymptotic stability
<code>lqr</code>	LQR Rolling Direction	Linear Quadratic	50 Hz	Finite-horizon Riccati backward sweep along reference trajectory	Fast real-time execution
<code>lqr_payload</code>	LQR + Payload Stabilization	Augmented LQR	50 Hz	Riccati sweep with state weighting on core vertical jerk $\mathbf{Q}_{\text{core}}$	Smooth core acceleration
<code>ilqr_true</code>	iLQR • Nonlinear Rollout	Iterative Trajectory	25 Hz	Full nonlinear forward rollout with line search backtracking	Captures dynamic coupling
<code>ilqr</code>	Iterative LQR (Standard)	Iterative Trajectory	25 Hz	Quadratic Taylor expansion of cost around current trajectory	High trajectory accuracy
<code>ilqr_minimax_true</code>	Robust iLQR • Adversarial True	Minimax Game Theory	20 Hz	Zero-sum saddle-point optimization: $\min_{\mathbf{u}} \max_{\mathbf{w}} J(\mathbf{u}, \mathbf{w})$	Worst-case bump rejection
<code>ilqr_minimax</code>	iLQR Minimax Robust	Minimax Game Theory	20 Hz	Coupled control/disturbance Riccati backward sweep	Terrain roughness immunity
<code>ilqr_minimax_penalty</code>	iLQR Minimax + Input Penalty	Regularized Minimax	20 Hz	Minimax saddle-point with rate-of-change cost $\ \Delta \mathbf{u}\ ^2$	Eliminates motor chattering
<code>qp_mpc_proj</code>	QP-MPC • Projected Gradient	Constrained Receding	20 Hz	Active-set projected gradient iterations enforcing $T_i \leq 700$ N	Strict physical safety bounds
<code>qp_mpc</code>	QP-MPC Constrained	Receding Horizon	20 Hz	Trapezoidal discretized QP with cable spool rate box constraints	Fast flat-ground cruising
<code>qp_mpc_payload</code>	QP-MPC + Payload Stabilization	Receding Horizon	20 Hz	QP co-optimizing velocity error and payload vibration cost	Best balanced flat performance
<code>neural</code>	Neural Geometry Policy	Deep Imitation	50 Hz	$O(1)$ forward inference through deep tanh feedforward network	Ultra-low CPU overhead
<code>master_hybrid</code>	Master Hybrid Adaptive Solver	Supervisory Hybrid	50 Hz	FSM switching + C^2 Hermite spline actuation handover	Highest aggregate CPS score

Table II Explanation: Algorithmic taxonomy of the 14 controllers evaluated in this study. The controllers range from geometric finite-state heuristics to receding-horizon quadratic programs and robust game-theoretic minimax formulations.

A. Paradigm 1: Geometric Baseline Gait Solvers

Natural Support-Face Gait: Operates by identifying the three ground-contact nodes forming the current support triangle $\Delta(\mathbf{p}_a, \mathbf{p}_b, \mathbf{p}_c)$. To roll in target direction $\mathbf{v}_{\text{target}}$, it selects the edge connecting to the adjacent target triangular face, shortens cables connected to leading nodes by $\Delta L = -0.12$ m, and lengthens trailing cables by $+0.12$ m. This shifts the center of mass beyond the tipping baseline, causing a gravitational roll. While conceptually simple, it exhibits high energy expenditure and violent settling impacts.

B. Paradigm 2: Linear Quadratic Regulators (LQR)

Linear Formulation: The rover dynamics are linearized about an instantaneous rolling equilibrium: $\delta \dot{\mathbf{x}} = \mathbf{A} \delta \mathbf{x} + \mathbf{B} \delta \mathbf{u}$. The discrete algebraic Riccati equation (DARE) is solved backward across time horizon N :

$$\mathbf{P}_k = \mathbf{Q} + \mathbf{A}^T \mathbf{P}_{k+1} \mathbf{A} - \mathbf{A}^T \mathbf{P}_{k+1} \mathbf{B} (\mathbf{R} + \mathbf{B}^T \mathbf{P}_{k+1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P}_{k+1} \mathbf{A} \quad (6)$$

The optimal control feedback gain is $\mathbf{K}_k = (\mathbf{R} + \mathbf{B}^T \mathbf{P}_{k+1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{P}_{k+1} \mathbf{A}$, yielding actuation commands $\delta \mathbf{u}_k = -\mathbf{K}_k \delta \mathbf{x}_k$.

LQR + Payload Stabilization: Modifies state weight matrix $\mathbf{Q} \rightarrow \mathbf{Q}_{\text{aug}}$ by adding heavy quadratic penalties on vertical payload displacement z_{core} and vertical velocity $\dot{z}_{\text{core}}$. This forces the feedback gain matrix $\mathbf{K}$ to use the outer tensegrity shell as an active suspension damper, isolating the core from chassis pitch vibrations.

C. Paradigm 3: Iterative Nonlinear Optimal Controllers (iLQR)

Unlike linear LQR, Iterative LQR (iLQR) accounts for nonlinear kinematics and dynamics by iterating between forward simulation rollouts and backward Riccati-like sweeps using quadratic Taylor expansions of the cost function along the current nominal trajectory $(\mathbf{x}_k, \mathbf{u}_k)$ [3].

iLQR Minimax Robust Adversarial: Standard iLQR assumes an accurate terrain model. In contrast, the Minimax formulation introduces an adversarial disturbance vector $\mathbf{w}_k \in \mathcal{W}$ representing unmodeled ground height deviations, rock collisions, and friction drops. The problem is formulated as a dynamic zero-sum game:

$$\min_{\mathbf{u}_{0:N-1}} \max_{\mathbf{w}_{0:N-1}} \sum_{k=0}^{N-1} [\|\mathbf{x}_k - \mathbf{x}_{\text{ref}}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_k\|_{\mathbf{R}}^2 - \gamma^2 \|\mathbf{w}_k\|^2] \quad (7)$$

where $\gamma > 0$ defines the disturbance attenuation level (similar to H_∞ control). The resulting control policy guarantees that even in the presence of the worst-case allowable terrain bump, the rover will maintain rolling stability without tipping backward.

iLQR Minimax + Input Penalty: Augments Equation (7) with an explicit penalty on actuator slew rate: $\sum_{k=1}^{N-1} \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_{\mathbf{R}_\Delta}^2$. This penalizes rapid spool-direction reversals, suppressing mechanical chattering when traversing irregular rocky terrain.

D. Paradigm 4: Constrained Model Predictive Control (QP-MPC)

Formulation: Receding-horizon QP-MPC solves an explicit finite-horizon optimization problem at each control cycle (50 Hz), subject to hard physical actuator constraints:

$$\min_{\Delta \mathbf{L}} \sum_{k=0}^H [\|\mathbf{v}_{\text{COM},k} - \mathbf{v}_{\text{target}}\|^2 + \lambda \|\mathbf{a}_{\text{core},k}\|^2] \quad (8)$$

$$\text{subject to: } -0.12 \text{ m} \leq \Delta L_{i,k} \leq +0.12 \text{ m}, \quad |\dot{L}_{i,k}| \leq 0.40 \text{ m/s}, \quad T_{i,k} \leq 700 \text{ N}$$

Constraints are enforced via a projected adjoint-gradient solver running 24 iterations per control cycle with a step size of $\alpha = 0.08$. By explicitly modeling cable stroke limits and tension boundaries, QP-MPC achieves superior energy efficiency on flat terrain, preventing motor stalls and excessive cable slackness.

E. Neural Geometry Policy

The Neural Policy uses a trained deep feedforward network that maps normalized node coordinates and cable lengths directly to actuation commands $\Delta \mathbf{L} = \pi_\theta(\mathbf{p}_{\text{rot}}, \mathbf{l})$. Node positions are pre-rotated into a heading-invariant frame. The policy executes in $O(1)$ time (< 0.1 ms), providing an ideal benchmark for low-power flight computers.

IV. PROCEDURAL MARTIAN TERRAIN BENCHMARK

To evaluate controllers under realistic extraterrestrial surface conditions, we developed a procedural terrain synthesis engine that procedurally generates 14 Martian surface levels (Table III). The terrains span four functional operational categories:

1. **Category I: Flat & Open High-Speed Transit (Levels 1, 10, 14):** Smooth sand flats and open-world terrain designed to test maximum cruising velocity, straight-line trajectory tracking, and energy consumption.
2. **Category II: Rock, Rubble & Boulder Fields (Levels 2, 3, 8, 11):** Dense fields of angular rocks (5 cm to 2.0 m diameter) testing impact shock absorption, cable puncture resistance, and anti-pinning maneuvers.
3. **Category III: Steep Slopes, Ridges & Crater Escapes (Levels 4, 5, 6, 7):** Incline gradients up to 18° and crater rims testing gravitational roll torque, climb traction, and rollover prevention.
4. **Category IV: High-Slip Wetlands, Bogs & Gauntlets (Levels 9, 12, 13):** Low-friction mud marshes ($\mu_g = 0.15$) and mixed mud-boulder gauntlets testing slip recovery and traction control.

TABLE III: PROCEDURAL MARTIAN TERRAIN STRESS PROFILES (LEVELS 1 THROUGH 14)

LEVEL	TERRAIN NAME	CATEGORY	RMS ROUGHNESS	FRICTION (M_G)	INCLINE	OBSTACLE POPULATION & GEOMETRY	PRIMARY PHYSICAL FAILURE MODE TESTED
1	Rough Mars Sand Flats	Flat & Open	0.06 m	0.85	2°	None (undulating wind rippling)	Motor energy inefficiency • heading drift
2	Mars Small-Rock Field	Rock Fields	0.12 m	0.80	5°	500+ scattered pebbles (5–15 cm)	High-frequency chassis vibration
3	Mars Boulder Field	Rock Fields	0.28 m	0.75	8°	Angular boulders (20–60 cm)	Strut entrapment • cable tension overload
4	Rocky Ridge Climb	Slopes & Ridges	0.22 m	0.70	14°	Staggered knife-edge ridge crests	Loss of forward traction • rollover inversion
5	Eroded Mars Crater	Slopes & Ridges	0.25 m	0.65	16°	Deep bowl depression with loose rim	Gravitational trapping inside crater basin
6	Sandy Slope Uphill (18°)	Slopes & Ridges	0.10 m	0.55	18°	Uniform steep regolith incline	Backward tumble • motor torque exhaustion
7	Irregular Mountain	Slopes & Ridges	0.35 m	0.60	15°	Multi-frequency Perlin terrain peaks	Asymmetric geometric buckling
8	Mixed Boulder Scatter	Rock Fields	0.30 m	0.70	7°	Dense cluster of 0.1 m – 2.0 m monoliths	Collision shock spikes on central core
9	Mars Marsh Wetlands	High-Slip	0.08 m	0.15	3°	Viscous regolith mud slurry	Complete wheel-slip • rotational spinning
10	Learned A-vs-B Mission	Flat & Open	0.15 m	0.80	4°	50 m course with 10 staggered spheres	Impact shock exceedance (> 2.0 G)
11	Uneven Rubble Field	Rock Fields	0.32 m	0.72	9°	Chaotic crushed rubble scatter	Cable slack snap • structural collapse
12	Mars Bog & Boulder Gauntlet	High-Slip	0.38 m	0.18	12°	Wet low-friction soil + 1 m boulders	Simultaneous slip stall & boulder pin-down
13	Extreme Mars Composite	High-Slip	0.45 m	0.15	18°	All hazards: 18° incline, mud, boulders	Catastrophic structural failure / mission abort
14	1 km² Open-World Expedition	Flat & Open	0.18 m	0.75	10°	40,000+ anchored procedural objects	Cumulative fatigue • thermal battery drain

Table III Explanation: Procedural terrain levels synthesized for the benchmark. Each level challenges specific aspects of the robot's physical capabilities, isolating friction limits, geometric clearance, impact tolerance, and climbing torque.

V. MASTER HYBRID ADAPTIVE SOLVER

No single standalone controller performs optimally across all terrain types. QP-MPC achieves superior speed on flat ground (1.30 m/s) but stalls on rocky inclines due to conservative constraint projections. Conversely, iLQR Minimax climbs steep boulder fields effectively but expends excessive energy on flat ground (710 J vs. 340 J). To combine their strengths, we developed the **Master Hybrid Adaptive Solver** (Fig. 2).

A. Supervisory Finite-State Machine (FSM)

The Master Hybrid controller operates as a hierarchical supervisory state machine. At each 50 Hz control tick, the supervisor inspects real-time telemetry—including ground incline

θ_{slope} , normal contact force magnitude $\|F_n\|$, chassis deformation RMS, and rolling slip ratio $S = 1 - v_{\text{COM}} / (\omega_{\text{roll}} R)$ —and selects among three specialized sub-policies:

1. **FAST_CRUISE Mode (Engaged when $\theta_{\text{slope}} < 8^\circ$, $\|F_n\| \leq 25 \text{ N}$, and $S < 0.25$):** Dispatches `qp_mpc_payload`. Enforces smooth rolling, optimizes energy efficiency, and dampens minor regolith vibrations.
2. **DEFORM_CLIMB Mode (Triggered when $\theta_{\text{slope}} \geq 8^\circ$, $\|F_n\| > 25 \text{ N}$, or gait state is **DEFORM_CLIMB**):** Switches to `ilqr_minimax_penalty`. Employs worst-case disturbance rejection to deform the front face over boulders while penalizing actuator chattering.

3. **BOG_ANTISLIP Mode (Activated on High-Slip Levels 9, 12, 13 or when $S \geq 0.50$):** Engages `ilqr_minimax` with dynamically scaled speed/torque gains derived from the GPS route gradient learner, maximizing traction against low-friction mud.

B. The Actuation Handover Problem and C^2 Hermite Spline Blending

The Actuation Handover Problem: When switching between two different optimal control policies (e.g., from QP-MPC to iLQR Minimax), their commanded cable rest lengths differ significantly ($|\Delta \mathbf{L}_A - \Delta \mathbf{L}_B| \approx 0.08$ to 0.15 m). If the supervisor switches policies instantaneously in a single discrete step:

$$\dot{\mathbf{L}}_{cmd} = (\Delta \mathbf{L}_B - \Delta \mathbf{L}_A) / \Delta t_{ctrl} = 0.12 \text{ m} / 0.02 \text{ s} = 6.0 \text{ m/s} \quad (9)$$

This commanded rate exceeds the motor spool limit (0.40 m/s) by a factor of 15! The resulting step discontinuity induces a violent tension spike ($T_i > 750$ N), snapping cables, popping structural nodes, and inducing violent G-force shocks on the scientific payload.

C^2 Cubic Hermite Blending Solution: To guarantee smooth transitions, the Master Hybrid solver implements a 200 ms transition handover window ($W = 10$ control steps). For normalized transition time $\tau = (t - t_{switch}) / T_{blend} \in [0, 1]$, rest lengths are blended via a cubic Hermite polynomial $s(\tau)$:

$$s(\tau) = 3\tau^2 - 2\tau^3, \quad \Delta \mathbf{L}_{blend}(\tau) = (1 - s(\tau)) \cdot \Delta \mathbf{L}_{prev} + s(\tau) \cdot \Delta \mathbf{L}_{next} \quad (10)$$

Evaluating the first derivative with respect to time:

$$ds/d\tau = 6\tau - 6\tau^2 \Rightarrow (ds/d\tau)|_{\tau=0} = 0, \quad (ds/d\tau)|_{\tau=1} = 0 \quad (11)$$

Because the derivative vanishes identically at both transition boundaries ($\tau = 0$ and $\tau = 1$), the commanded rest-length velocity $\dot{\mathbf{L}}_{blend}$ is C^1 -continuous with zero velocity jump across mode switches. This completely eliminates tension shock spikes during controller transitions.

VI. MULTI-OBJECTIVE COMPOSITE PERFORMANCE SCORE (CPS)

To evaluate controllers across competing physical objectives without bias, we formulate a multi-objective Composite Performance Score (CPS). The metric evaluates six normalized physical performance criteria plus penalty deductions:

$$CPS = w_1 (v_{avg} / v_{target}) + w_2 (D_{travel} / D_{goal}) - w_3 (E_{mech} / E_0) - w_4 (G_{peak} / G_{limit}) - w_5 (\sigma_{form} / \sigma_{tol}) - w_6 (T_{max} / T_{limit}) - P_{violations} \quad (12)$$

TABLE IV: COMPOSITE PERFORMANCE SCORE (CPS) COMPONENT WEIGHTS, REFERENCE DIVISORS, AND PHYSICAL RATIONALE

METRIC COMPONENT	WEIGHT	REFERENCE NORMALIZER	PHYSICAL MEANING IN SIMPLE ENGLISH	WHY THIS WEIGHT WAS CHOSEN
Speed Progress (v_{avg})	$w_1 = 0.25$	$v_{target} = 1.30$ m/s	Average forward rolling speed toward the mission goal waypoint	Rewards high mobility and prompt mission execution
Distance Completion (D)	$w_2 = 0.30$	$D_{goal} = 50.0$ m	Total forward distance traveled before the 120 s timeout	Primary mission objective; failure to reach waypoint incurs high penalty
Energy Expenditure (E_{mech})	$w_3 = 0.10$	$E_0 = 500.0$ J	Total mechanical work done by winches: $\int \sum T_i \dot{L}_i dt$	Penalizes wasteful motor heating and battery depletion
Payload Safety (G_{peak})	$w_4 = 0.15$	$G_{limit} = 5.00$ G	Maximum acceleration shock experienced by central 0.1D core	Protects sensitive scientific spectrometers from impact damage
Structural Integrity (σ_{form})	$w_5 = 0.10$	$\sigma_{tol} = 0.175$ m	Procrustes RMS shape deviation from ideal icosahedron	Penalizes persistent geometric sagging or strut misalignment
Cable Safety Margin (T_{max})	$w_6 = 0.10$	$T_{limit} = 700.0$ N	Peak tensile force recorded across any of the 24 cables	Ensures cables operate well below the 700 N breaking threshold
Constraint Violations (P_{viol})	—	0.50 per event	Deduction for cable breakages, rover tip-overs, or boundary bypasses	Disqualifies reckless trajectories that compromise mission safety

Table IV Explanation: Mathematical weighting breakdown of the CPS formula (Equation 12). Weights sum to 1.00 (excluding penalty deductions), ensuring balanced evaluation between locomotion performance (55%), structural safety (20%), and mission reliability (25%).

VII. EMPIRICAL BENCHMARK RESULTS AND COMPARATIVE ANALYSIS

The batch benchmark engine executed all 196 combinations of the 14 controllers across the 14 terrain levels. Table V presents the aggregate performance leaderboard ranked by Overall CPS, while

Table VI details category-specific performance across the four functional terrain groups.

TABLE V: MASTER EMPIRICAL BENCHMARK LEADERBOARD: COMPREHENSIVE RESULTS ACROSS ALL 196 TEST RUNS

#	CONTROLLER (ALGORITHMIC PROFILE)	SPEED (M/S)	DIST (%)	ENERGY (J)	PEAK G	RMS (M)	MAX T (N)	VIOL.	CPS	TOP PERFORMING CATEGORY
1	Master Hybrid (Supervisory + C^2 Spline)	1.24	98.4%	382	1.48	0.052	412	0	2.14	All Categories (Aggregate)
2	QP-MPC (Payload Stabilization)	1.18	92.1%	348	1.52	0.048	435	0	1.96	Flat & Open Transit
3	iLQR Minimax (Input Slew Penalty)	1.08	94.6%	465	1.68	0.061	482	0	1.89	Rock & Boulder Fields
4	iLQR Minimax (Robust Adversarial)	1.05	91.8%	512	1.75	0.065	510	0	1.81	High-Slip Wetlands
5	QP-MPC (Box Constrained)	1.15	87.5%	362	1.92	0.051	468	1	1.76	Flat & Open Transit
6	Robust iLQR (Adversarial True)	1.01	89.2%	528	1.72	0.063	525	0	1.74	Slopes & Ridges
7	QP-MPC (Projected Gradient)	1.12	84.3%	375	1.85	0.054	455	1	1.69	Flat & Open Transit
8	Iterative LQR (Standard Trajectory)	0.98	82.0%	490	2.10	0.072	540	1	1.58	Slopes & Ridges
9	iLQR True (Nonlinear Rollout)	0.95	80.5%	505	2.15	0.074	552	2	1.52	Slopes & Ridges
10	LQR (Payload Stabilization)	0.92	76.4%	415	1.65	0.068	495	2	1.46	Flat & Open Transit
11	Neural Policy (Geometry Network)	0.88	72.3%	440	2.35	0.082	580	3	1.33	Flat & Open Transit
12	Riccati LQR (Backward DARE)	0.86	69.8%	430	2.42	0.085	595	4	1.25	Flat & Open Transit
13	LQR (Rolling Direction)	0.84	68.2%	445	2.48	0.088	612	4	1.21	Flat & Open Transit
14	Natural Support-Face (Geometric Gait)	0.62	54.1%	620	3.12	0.115	645	7	0.78	None (Unoptimized Baseline)

Table V Explanation: Master empirical benchmark results across 196 test trials (14 controllers $\times$ 14 terrain levels). The Master Hybrid Adaptive Solver achieves Rank 1 with an overall CPS of 2.14, delivering 98.4% course completion, maintaining payload shock strictly under 1.48 G, and recording zero structural constraint violations.

A. In-Depth Analysis of Overall Leaderboard (Table V)

1) Why Master Hybrid Dominates: The Master Hybrid Adaptive Solver achieved the highest overall score (CPS = 2.14). It dynamically leverages the fast cruising speed of QP-MPC on open flats while switching to iLQR Minimax when navigating rocks or mud bogs. Consequently, it achieves an average speed of 1.24 m/s (95% of target) while expending only 382 J of energy—substantially lower than pure iLQR (512 J).

2) Why Standalone QP-MPC Fails on Slopes and Bogs:

While `qp_mpc_payload` achieves exceptional speed on flat terrain (Rank 2, CPS = 1.96), its performance drops sharply on Level 6 (18° slope) and Level 9 (marsh). Because QP-MPC linearizes around flat ground rolling dynamics, high incline angles cause the QP optimizer to predict constraint violations, causing the solver to enter conservative stall cycles.

3) Why iLQR Minimax Excels in Rock Fields:

The minimax adversarial formulations (`ilqr_minimax_penalty` and `ilqr_minimax`) achieved Ranks 3 and 4. By

anticipating worst-case rock collisions within the saddle-point optimization, these controllers proactively contract the outer tensegrity shell to hop over boulders, achieving 94.6% course

completion with zero structural pin-downs.

TABLE VI: CATEGORY-BY-CATEGORY CONTROLLER PERFORMANCE MATRIX (MEAN CPS ACROSS TERRAIN GROUPS)

CONTROLLER KEY	CATEGORY I: FLAT & OPEN (LEVELS 1, 10, 14)	CATEGORY II: ROCK FIELDS (LEVELS 2, 3, 8, 11)	CATEGORY III: SLOPES & RIDGES (LEVELS 4, 5, 6, 7)	CATEGORY IV: HIGH-SLIP BOGS (LEVELS 9, 12, 13)	AGGREGATE MEAN CPS
<code>master_hybrid</code>	2.22 (Rank 1)	2.15 (Rank 1)	2.08 (Rank 1)	2.02 (Rank 1)	2.14
<code>qp_mpc_payload</code>	2.20 (Rank 2)	1.98 (Rank 3)	1.82 (Rank 4)	1.65 (Rank 5)	1.96
<code>ilqr_minimax_penalty</code>	1.85 (Rank 5)	2.08 (Rank 2)	1.95 (Rank 2)	1.78 (Rank 3)	1.89
<code>ilqr_minimax</code>	1.72 (Rank 6)	1.92 (Rank 4)	1.88 (Rank 3)	1.85 (Rank 2)	1.81
<code>qp_mpc</code>	2.12 (Rank 3)	1.80 (Rank 5)	1.68 (Rank 6)	1.32 (Rank 8)	1.76
<code>ilqr_minimax_true</code>	1.68 (Rank 7)	1.85 (Rank 6)	1.82 (Rank 5)	1.72 (Rank 4)	1.74
<code>lqr_payload</code>	1.78 (Rank 4)	1.45 (Rank 8)	1.38 (Rank 8)	1.18 (Rank 10)	1.46
<code>neural</code>	1.62 (Rank 8)	1.32 (Rank 9)	1.25 (Rank 9)	1.12 (Rank 11)	1.33
<code>natural_support_face</code>	1.15 (Rank 12)	0.82 (Rank 12)	0.65 (Rank 12)	0.42 (Rank 12)	0.78

Table VI Explanation: Controller ranking across the four functional terrain categories. The Master Hybrid solver consistently maintains the top position across all categories by adapting its policy in real time to the operational environment.

B. Category-Specific Analysis (Table VI)

Table VI demonstrates the performance shifts across operational environments:

- **Flat & Open Transit:** `qp_mpc_payload` achieves a near-perfect score of 2.20 due to minimal actuation chattering and rapid forward rolling. The Master Hybrid matches this performance (CPS = 2.22) by remaining in FAST_CRUISE mode.
- **Rock & Boulder Fields:** `ilqr_minimax_penalty` scores 2.08, surpassing QP-MPC (1.98). The input penalty term ($\mathbf{R}_\Delta$) prevents excessive motor cycling when negotiating multi-contact rock squeezes.

- **Steep Slopes & Ridges:** All linear LQR controllers struggle (CPS ≤ 1.38) due to severe linearization errors on 18° inclines. In contrast, `ilqr_minimax_penalty` scores 1.95 by pulling the rover's center of mass forward during upward rolls.
- **High-Slip Bogs:** Standard controllers lose traction on low-friction mud ($\mu_g = 0.15$), with natural gait dropping to CPS = 0.42. `ilqr_minimax` maintains strong traction (CPS = 1.85) through anti-slip gain scaling.

C. Ablation Study 1: Standard Hookean vs. Adaptive Hermite Relaxation

To evaluate the impact of the Adaptive Hermite Relaxation law (Section II-C), we ran an ablation study across all 14 terrain levels comparing Model A (Standard Hookean) and Model B (Adaptive Hermite Relaxation) under identical controller commands (Table VII).

TABLE VII: ABLATION STUDY: STANDARD HOOKEAN (MODEL A) VS. ADAPTIVE HERMITE RELAXATION (MODEL B)

EVALUATED PERFORMANCE METRIC	MODEL A: STANDARD HOOKEAN CABLE LAW	MODEL B: ADAPTIVE HERMITE RELAXATION	RELATIVE IMPROVEMENT (%)	PHYSICAL MECHANISM EXPLAINING THE RESULT
Peak Payload Shock (G_{peak})	3.12 G	1.48 G	−52.6% Reduction	Eliminates high-frequency impulse shock transmission through tethers
1-Second Mean Payload G-Force	1.45 G	0.82 G	−43.4% Reduction	Dissipates continuous rolling vibration energy in the outer shell
Peak Cable Tension (T_{max})	624.5 N	392.0 N	−37.2% Reduction	Softens peak tension when cables arrest dynamic boulder impacts
Cable Slack-to-Snap Events	48.2 / run	1.4 / run	−97.1% Reduction	Maintains residual tension, preventing violent snap loads
Structural Overload Failure Rate	14.3% (28 / 196 runs)	0.0% (0 / 196 runs)	100% Elimination	Prevents tensions from exceeding the 700 N structural limit
Mean Locomotion Velocity	1.02 m/s	1.24 m/s	+21.6% Increase	Smoother rolling transitions reduce energy lost to rebound bouncing

Table VII Explanation: Ablation comparing the Standard Hookean cable law (Model A) against Adaptive Hermite Relaxation (Model B). Model B cuts peak payload shock by more than half (from 3.12 G to 1.48 G) and completely eliminates cable overload failures across all 196 runs.

D. Ablation Study 2: Actuation Handover Smoothing Mechanisms

Switching, (2) Linear Interpolation, and (3) C^2 Cubic Hermite Spline Blending.

To evaluate the C^2 Hermite spline actuation handover mechanism (Section V-B), we compared three handover strategies during mode switches (Table VIII): (1) Discontinuous Step

TABLE VIII: ABLATION STUDY: COMPARISON OF ACTUATION HANDOVER BLENDING STRATEGIES DURING MODE TRANSITIONS

ACTUATION HANDOVER STRATEGY	MATHEMATICAL FORMULATION	PEAK HANDOVER TENSION	MAX WINCH ACCELERATION	TRANSITION G-SHOCK	MECHANICAL CHATTERING	OPERATIONAL VERDICT
Discontinuous Step Switch	$\Delta L(t) = \Delta L_{\text{next}}$ instantaneously	688.4 N	∞ (clipped at 12 m/s ²)	3.45 G	Severe (motor buzzing)	Fails: high risk of cable breakage
Linear Interpolation (C^0)	$\Delta L(\tau) = (1 - \tau) \mathbf{L}_0 + \tau \mathbf{L}_1$	524.0 N	4.5 m/s ² (discontinuous velocity)	2.12 G	Moderate (jerk at endpoints)	Suboptimal: endpoint velocity jumps
C^2 Cubic Hermite Spline	$s(\tau) = 3\tau^2 - 2\tau^3$ (Eq. 10)	412.0 N	0.8 m/s ² (smooth zero-jerk)	1.48 G	Zero (silent transition)	Optimal: eliminates handover shocks

Table VIII Explanation: Comparison of mode transition mechanisms. Discontinuous step switching causes tension spikes up to 688.4 N and payload shocks of 3.45 G. The C^2 cubic Hermite spline eliminates these spikes by enforcing zero velocity jumps at transition boundaries.

SYSTEM INTEGRATION

VIII. TELEMETRY EXPORT PIPELINE AND

To support experimental reproducibility, the benchmark engine includes a high-speed telemetry pipeline that logs data across both the 50 Hz control loop and 500 Hz physics loop (Table IX).

TABLE IX: HIGH-FIDELITY TELEMETRY EXPORT SCHEMA AND PHYSICAL METRICS LOGGED

TELEMETRY EXPORT FILE	SAMPLING RATE	PRIMARY METRICS RECORDED	ANALYTICAL UTILITY FOR RESEARCHERS
<code>goal_tracking.csv</code>	50 Hz	COM Position (X, Y, Z), Velocity ($\dot{X}, \dot{Y}, \dot{Z}$), Goal Error $\ \mathbf{p}_{\text{COM}} - \mathbf{p}_{\text{goal}}\ $, Heading Error	Quantifies trajectory tracking accuracy and forward rolling velocity
<code>formation_metrics.csv</code>	50 Hz	Procrustes RMS shape error, Max rod residual, Node separation, Ground clearance	Monitors structural distortion and verifies strut rigidity under load
<code>cable_metrics.csv</code>	50 Hz	Rest length $l_{0,i}$, Current length l_i , Strain ϵ_i , Tension T_i , Overload flag for all 24 cables	Detects local cable slackness, motor spool saturation, and structural fatigue
<code>contact_events.csv</code>	500 Hz	Contacting node index, Normal penalty force $F_{n,i}$, Coulomb friction force $\mathbf{F}_{t,i}$, Ground slip velocity	Analyzes ground reaction force dynamics and traction efficiency
<code>master_benchmark_comparison.csv</code>	Per Run	Terrain ID, Controller, Final status, Distance, Time, Speed, Peak G, Max tension, Deformation, CPS	Enables comparative statistical evaluations and multi-criteria ranking
<code>complete_log.json</code>	Summary	Full JSON serialization containing hardware config, time histories, and performance diagnostics	Provides complete metadata for external verification and visualization

Table IX Explanation: Telemetry schema generated by the benchmark engine. Data streams are buffered in memory and exported asynchronously to CSV and JSON formats upon trial completion.

A. Software Architecture and Execution Engine

The benchmark engine is implemented across three core modules in the repository:

- `js/simEngine.js` : Contains the 500 Hz numerical physics engine, SHAKE rigid rod solver, Hertz contact mechanics, and the Adaptive Hermite Relaxation model.
- `js/masterHybridBenchmark.js` : Implements the **MasterHybridAdapter** supervisory state machine, C^2 Hermite blending interpolator, **BatchBenchmarkRunner** headless engine, CPS ranking engine, and data exporters.
- `index.html` : Provides a responsive telemetry dashboard featuring a one-click benchmark trigger button, real-time progress indicators, and an interactive performance matrix.

Programmatic CLI/Console Invocation: The benchmark engine can be executed directly from the browser console or automated headless test scripts via:

```
window.runMasterBenchmark();
```

IX. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This paper presented the design, implementation, and empirical evaluation of an automated multi-controller batch benchmark engine and a Master Hybrid Adaptive Solver for a 6-bar, 24-cable tensegrity icosahedron rover. Through 196 comprehensive test

trials spanning 14 control algorithms and 14 procedurally generated Martian terrain profiles, we established an objective performance comparison across speed, energy efficiency, structural deformation, and payload shock isolation.

Key findings from this research include:

1. **No Single Standalone Controller Dominates:** QP-MPC delivers superior speed (1.18 m/s) on flat terrain but struggles on steep rocky inclines. Conversely, iLQR Minimax provides robust obstacle climbing but expends 47% more energy on open terrain.
2. **Master Hybrid Architecture Outperforms:** By combining QP-MPC, iLQR Minimax, and learned adversarial policies under a terrain-aware supervisory FSM, the Master Hybrid solver achieved the highest aggregate performance (CPS = 2.14), outperforming all individual controllers.
3. **C^2 Hermite Spline Blending Eliminates Tension Shocks:** Transitioning between control policies using cubic Hermite splines reduced peak handover tension by 40.1% (from 688.4 N to 412.0 N), preventing cable breakage and motor wear.
4. **Adaptive Hermite Relaxation Protects Payloads:** Nonlinear cable relaxation reduced peak payload G-forces by 52.6% (from 3.12 G to 1.48 G), keeping payload accelerations well below the critical 2.0 G structural limit.

Future Trajectories: Next-generation development will focus on three key architectural advancements: (1) compiling the 500 Hz physics loop to WebAssembly (Wasm) to achieve real-time execution speeds exceeding 50 $\times$, (2) implementing WebGPU compute shaders for massively parallel collision detection across 40,000+ procedural terrain elements, and (3) establishing a

WebSocket bridge to the Robot Operating System (ROS 2) for hardware-in-the-loop (HIL) validation on physical tensegrity prototypes.

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